

Aging in Atlanta and Key Amenities

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Objective / Problem Statement

- Atlanta has a growing senior population. In one of the country's most sprawling cities, how accessible are their basic needs?
- To create an index that evaluates the accessibility of essential services – as public transit, grocery stores, and medical care – for seniors living in affordable housing in Atlanta.
- To identify areas where seniors face significant barriers in accessing these services and find exploratory solutions or further research questions that promote walkability, connectivity, and overall quality of life within affordable housing communities.

Key Facilities & Weighting

MARTA Heavy Rail – 25%

Allows for access to essential services like healthcare, grocery stores, and social activities, fostering independence and reducing isolation. Reliable transit options also alleviate financial strain by minimizing the need for personal vehicles and repairs.

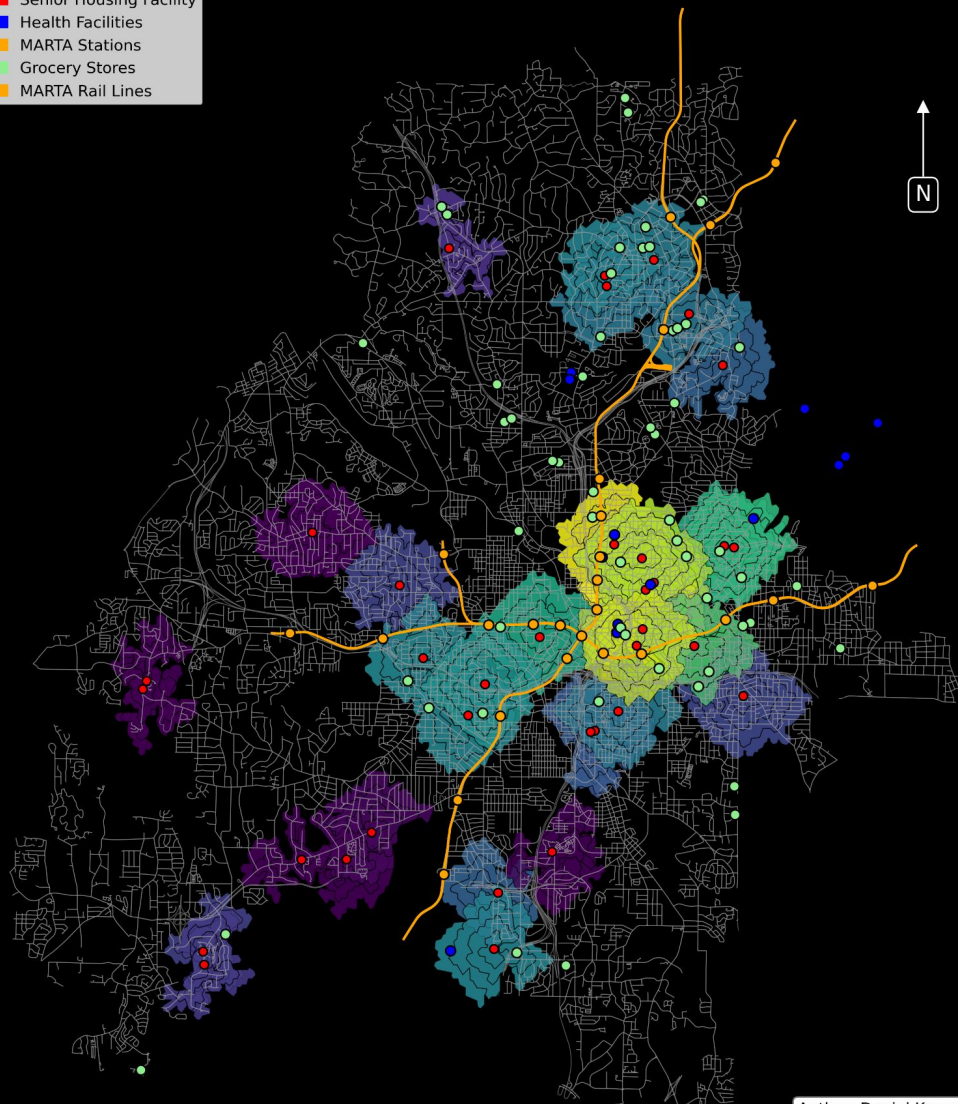
Grocery Stores – 30%

Ensures access to fresh, nutritious food, which is essential for maintaining health and managing chronic conditions. Nearby stores also reduce the burden of transportation costs and physical strain, promoting greater food security and independence.

Health Facilities – 45%

Crucial for timely access to medical care, which is essential for managing chronic conditions and maintaining overall well-being. Nearby healthcare services reduce transportation challenges and costs, fostering better health outcomes and independence.

Composite Accessibility Score Across Facilities



Mean Composite Accessibility Score: 0.36
Median Composite Accessibility Score: 0.37

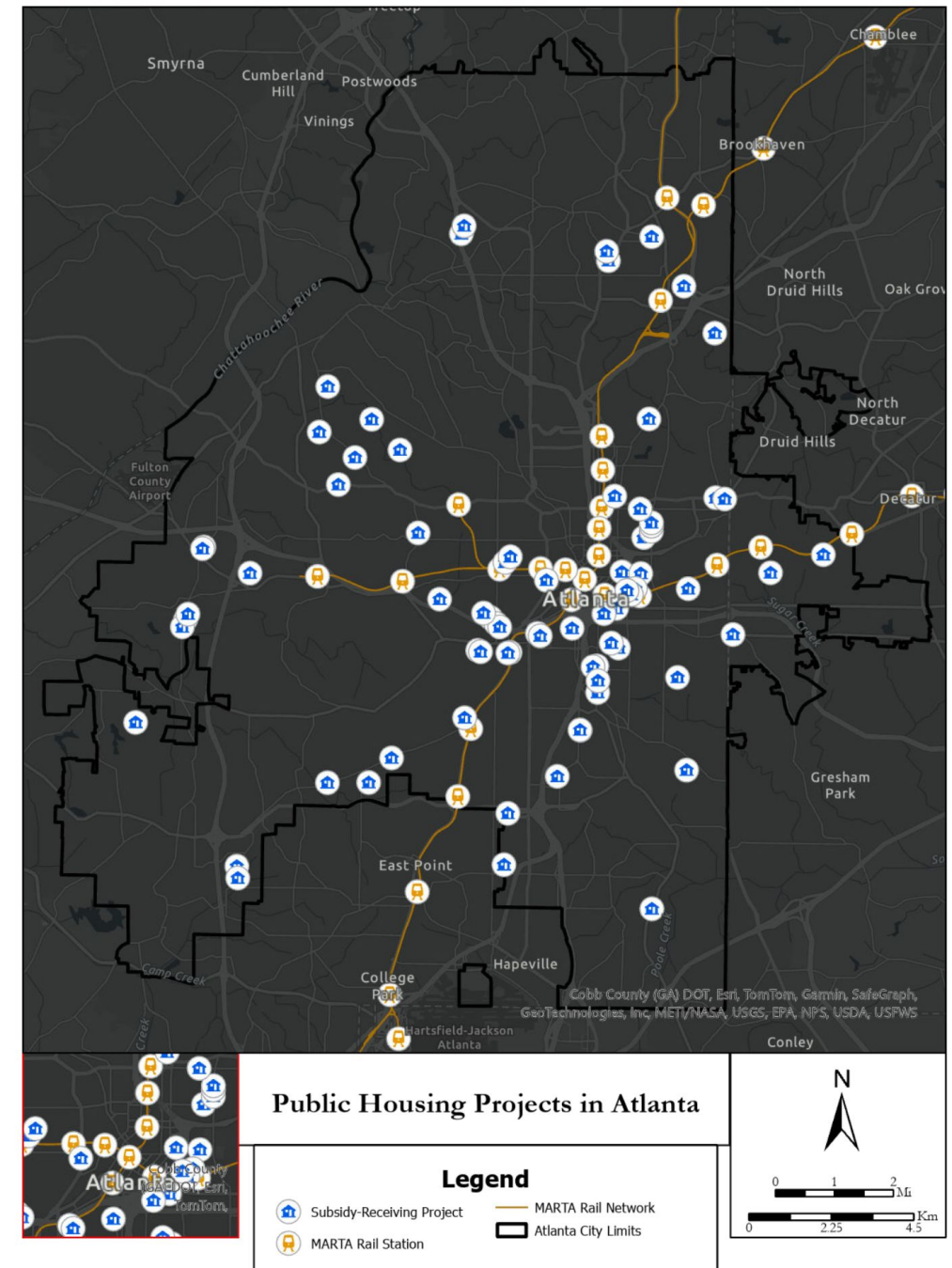
Author: Daniel Krasner
Date: 2024-11-25

Findings

- Results of the three-variable index indicate Midtown / Old Fourth Ward area affordable housing for seniors has highest accessibility to these features.
- This is the Central / North Central Corridor, slightly East, where land values are high but there's also walkability and transit options.
- Seems fairly predictable, but normalized composite scores are below 50%, which means there's significant outliers pulling values down comparative to the "more accessible" areas.

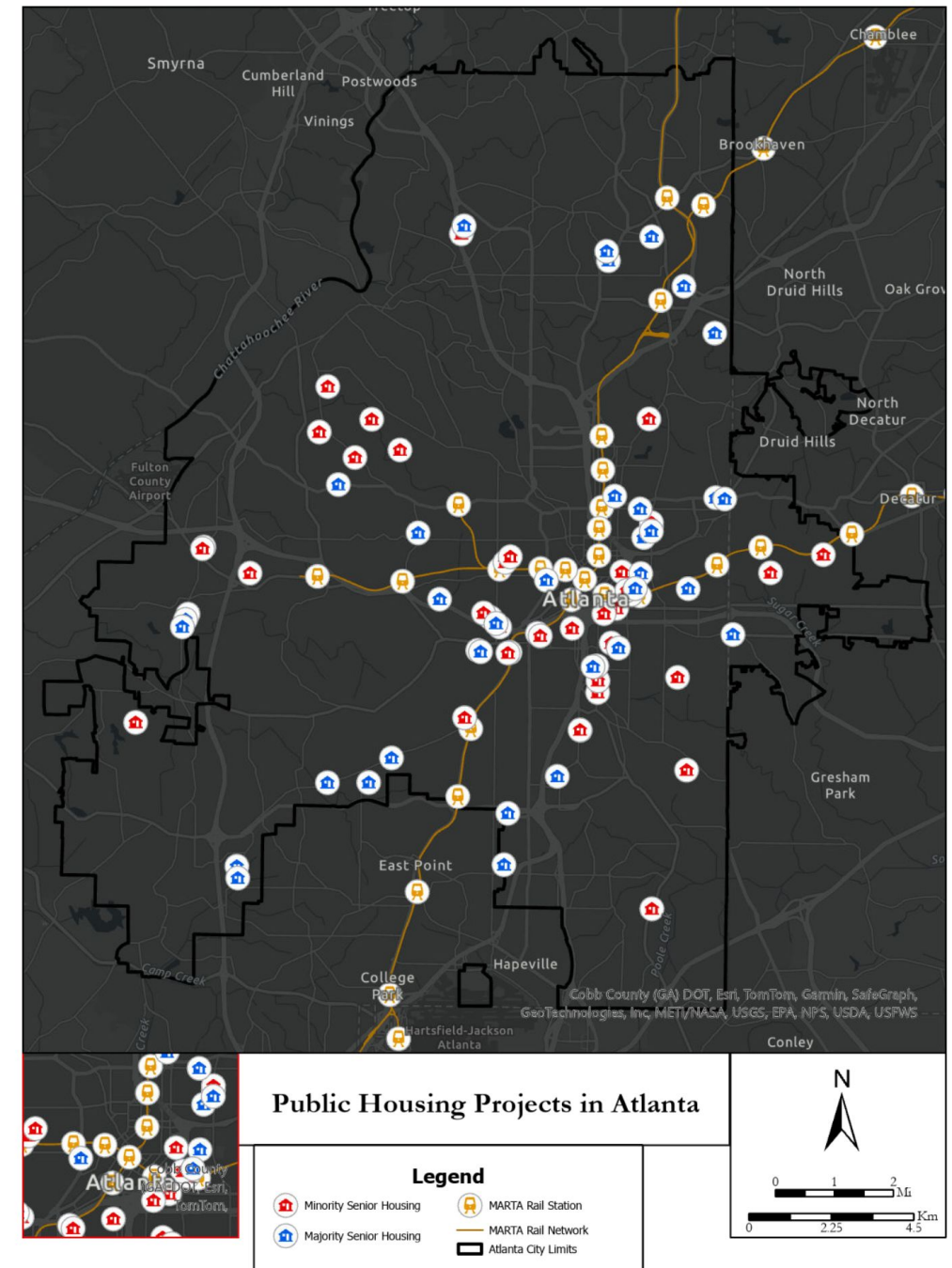
There are 89 social housing facilities in ATL

- Blue facilities are social housing while orange is MARTA heavy rail.
- Source: U.S. Department of Housing and Urban Development (HUD) website.
- Dataset: Assisted Housing: National and Local from the Picture of Subsidized Households collection.
- Overview: Includes project-level data on public housing, tenant-based, and privately-owned, project-based housing assistance programs.



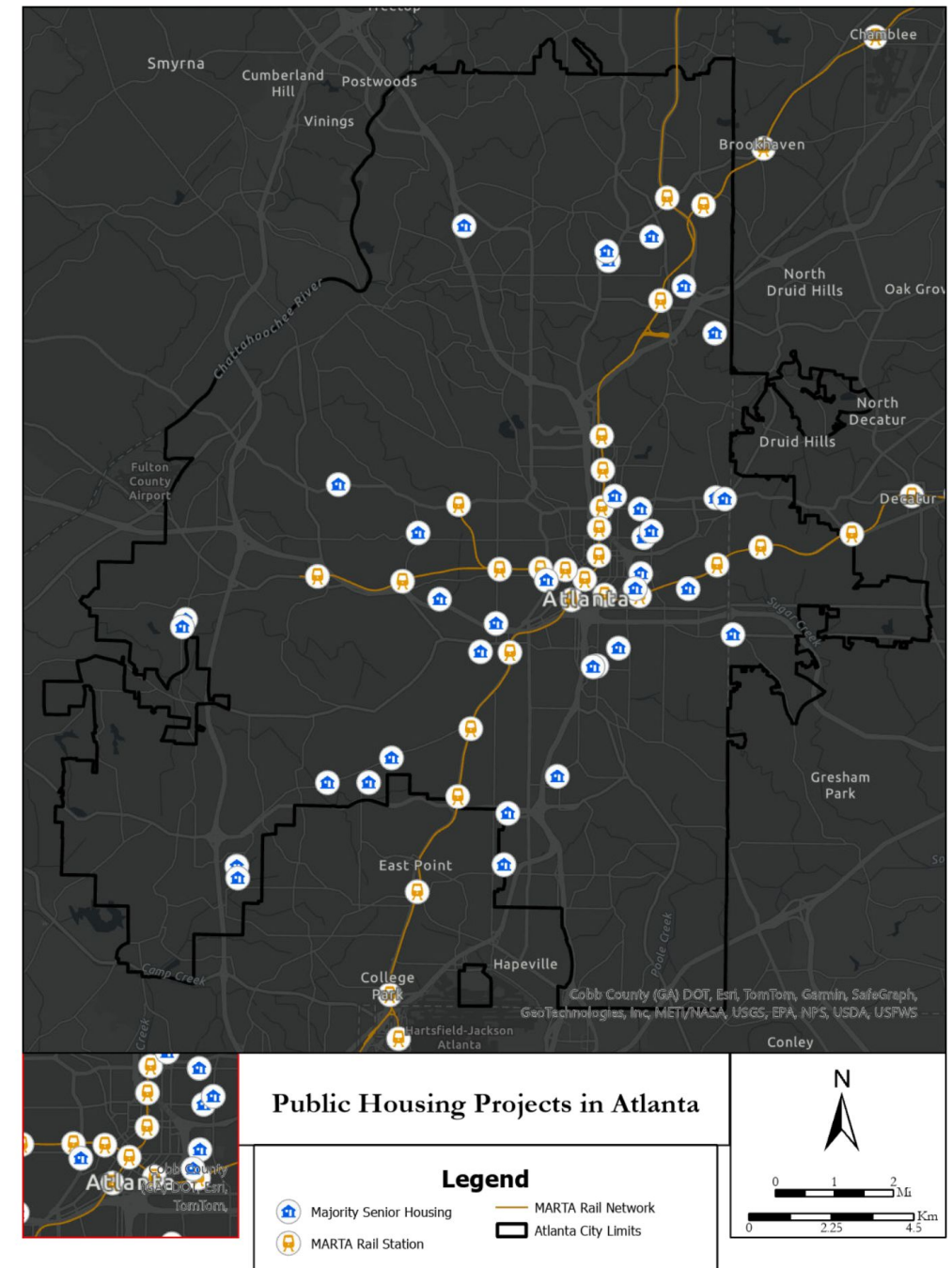
37 of those 89 are Considered Majority Senior

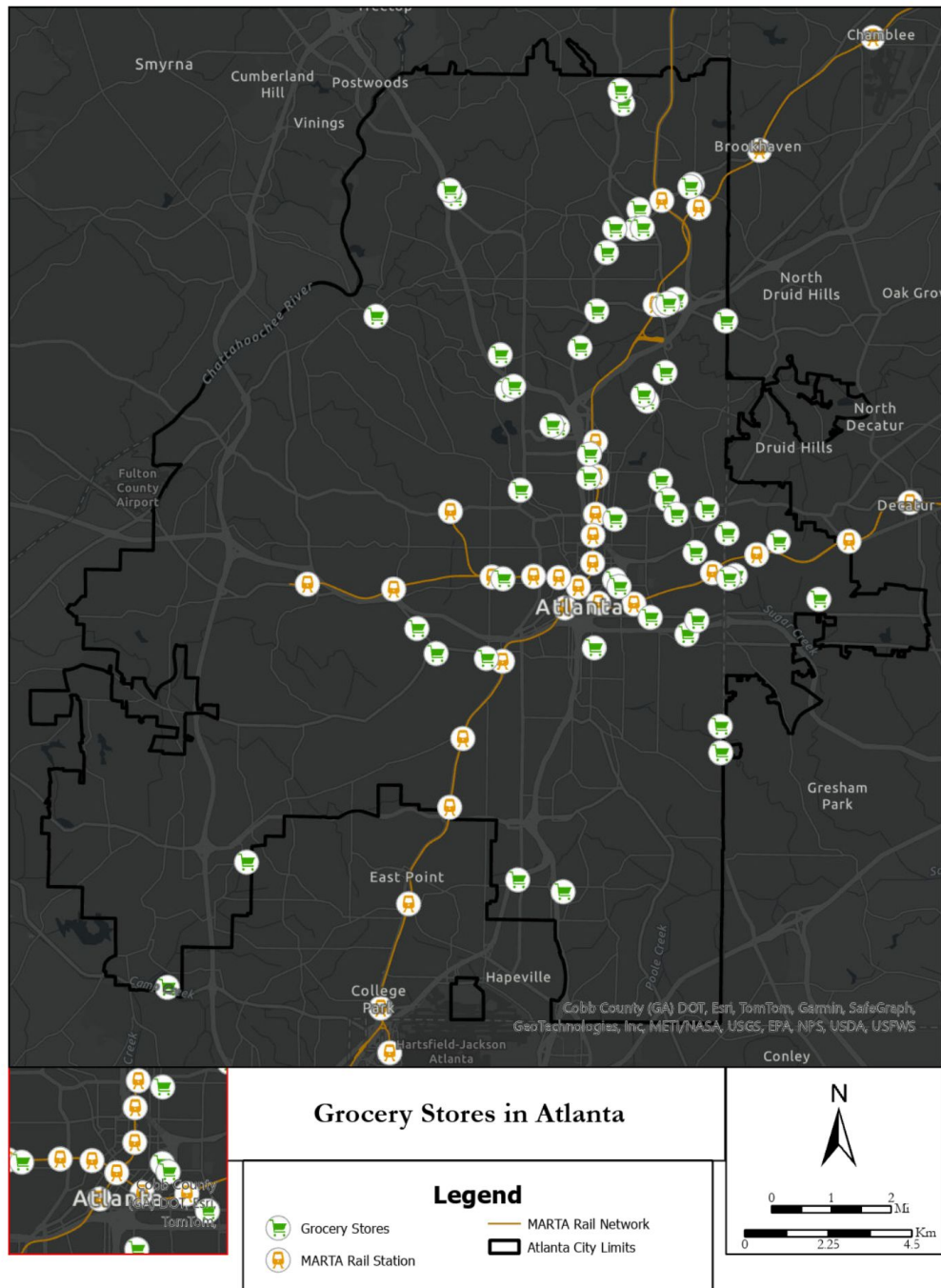
- Seniors in this case are defined as cases in which the older of the household head or spouse is **62 years of age or older**
- Filtered: For public housing developments where a majority (more than 50%) of the tenants are seniors
- On the map, the red facilities are all developments that **do not** fit this criteria



We will focus on the Remaining 41.6%

- For further analysis beyond this class, the rich demographic data of this data set may be explored
- Other variables include race, income, and HUD / rent expenditures as well as program / subsidy type.



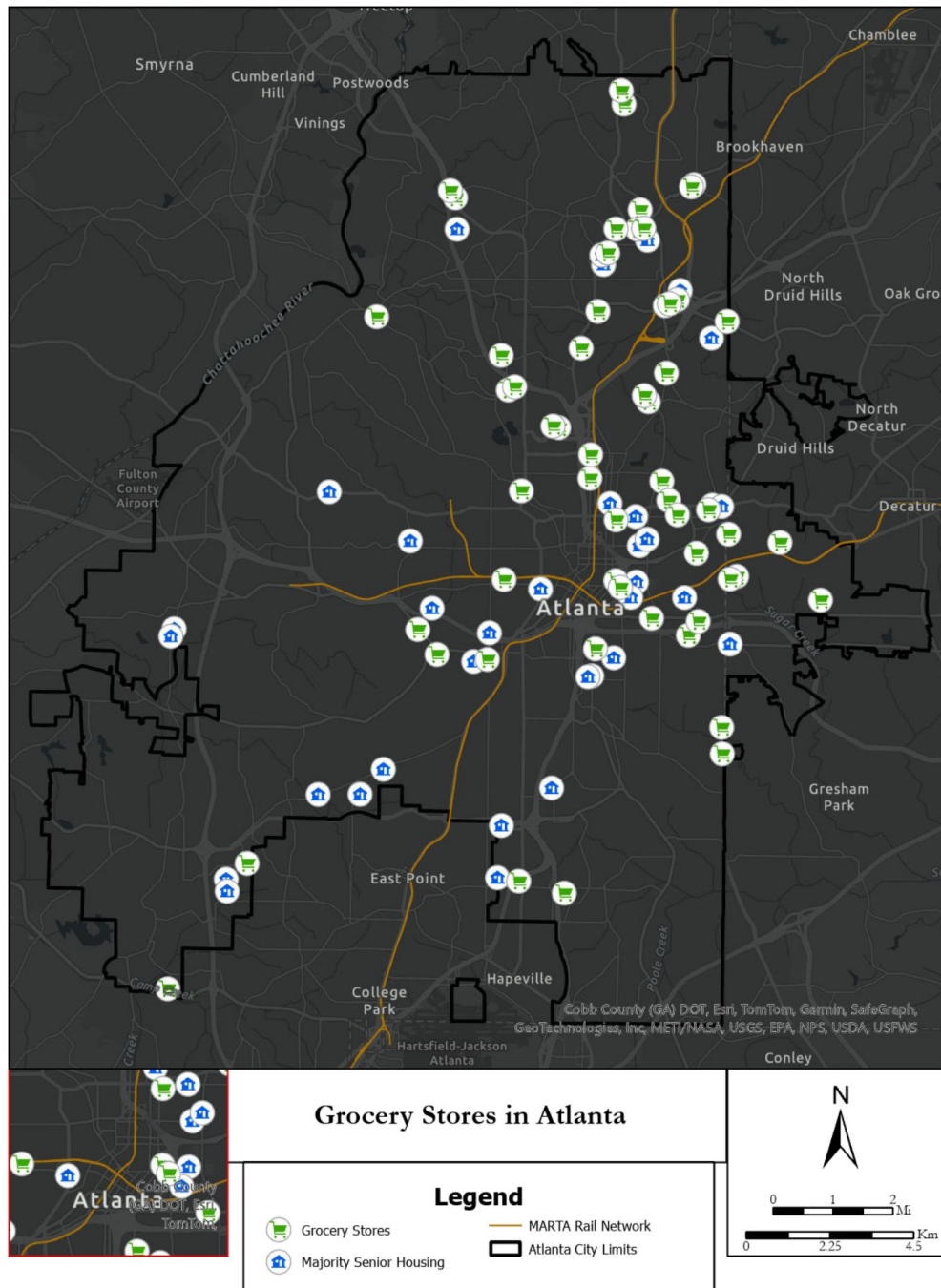


56 Grocery Stores are Included in the Analysis

- Green locations on the map
- Pulled from OSM with tags such as “convenience, supermarket, department store, green_grocer”
- Examples of stores captured in the data include, but are not limited to: Kroger, Target, Publix, Whole Foods Market, Trader Joe’s, Aldi, Walmart Neighborhood Market, The Fresh Market, Piggly Wiggly, Sprouts Farmers Market, Local and independent grocery stores

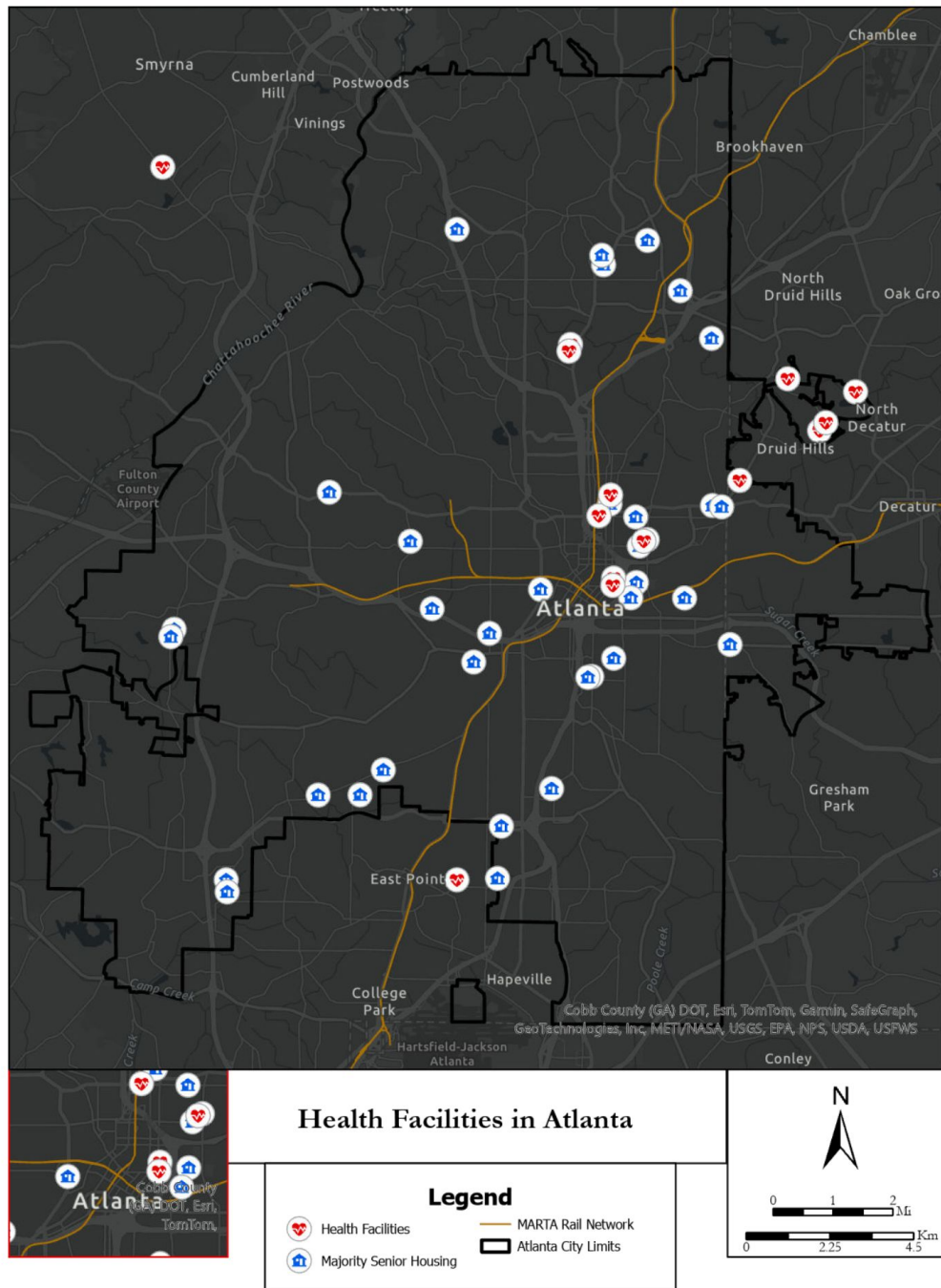
Where are Groceries Stores in Relation to Public Housing?

- Many convenience stores and food marts were excluded while some delis that sell produce were included.
- Tried to avoid anything that didn't have a website or seemed to only sell gas station-like snacks.
- This includes a lot of one-off facilities mostly on the southwest side, leaving this side of town as what might be considered food deserts while grocers are over concentrated in the northeast corridor.



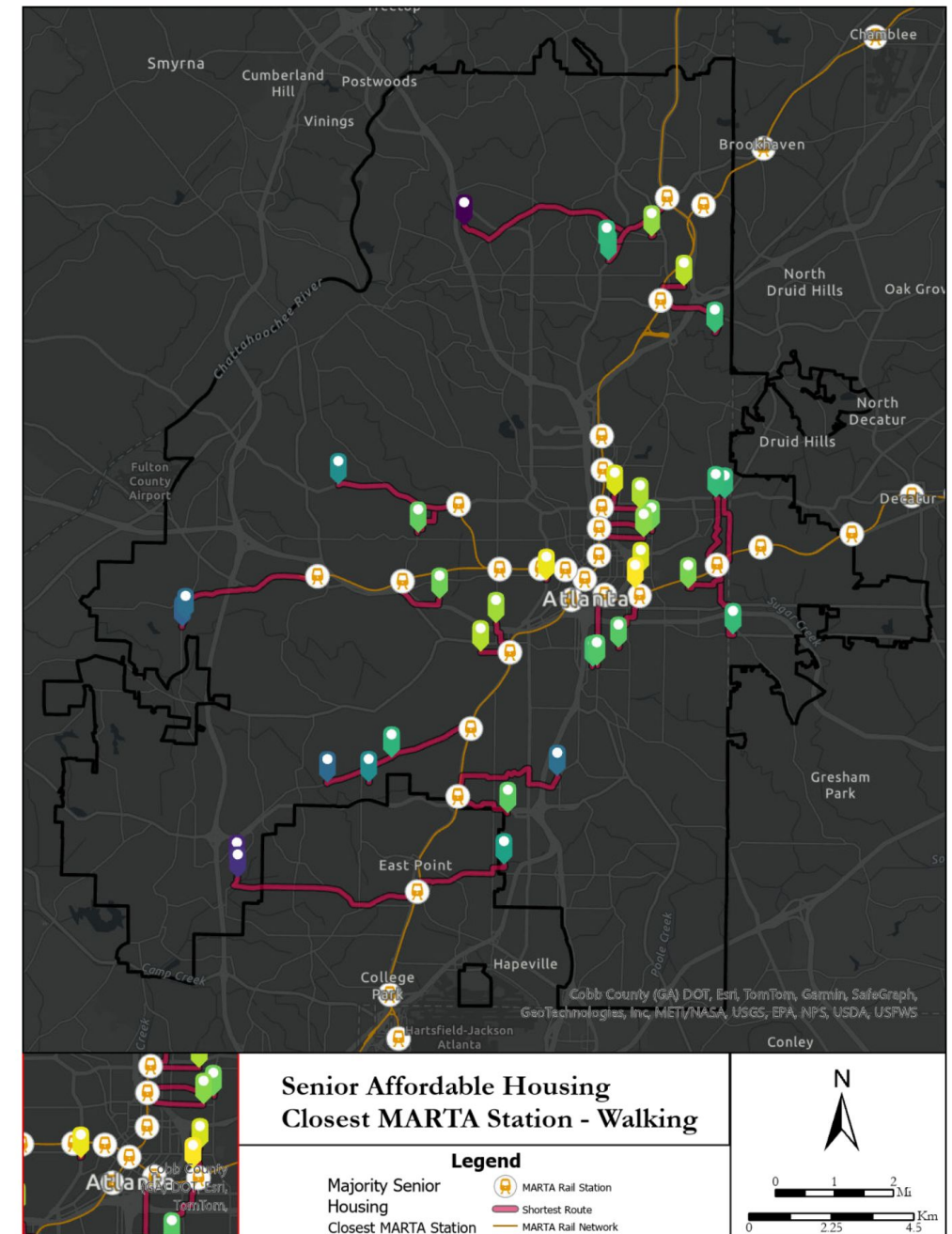
Where are Health Facilities in Relation to Public Housing?

- For the third variable, I pulled in the one Dr. He provided us for health facilities, which are marked in red.
- Hyper concentration in the middle corridor and northeast corner by Emory.
- Virtually nothing on the south or west side whatsoever.



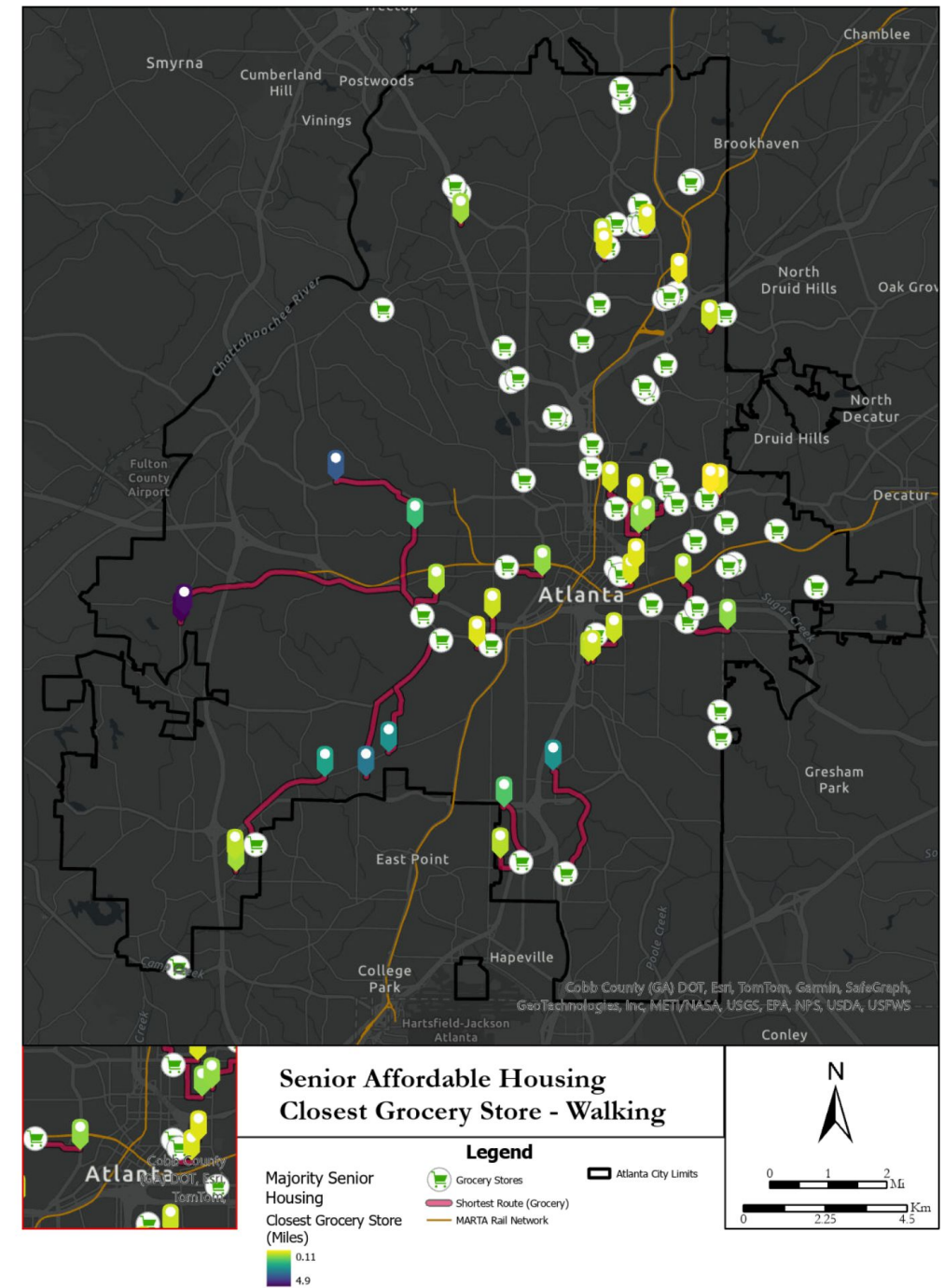
Closest Facility – MARTA

- Housing icons are now shaded based on distance from other facilities. Yellow is closest, purple is furthest.
- Pulled into ArcGIS Pro for shortest route analysis.
- Focused on “walk” network as a proxy for “calmer,” less speed/highway intensive routes, not necessarily assuming seniors would be walking these routes.
- Average distance to nearest = 1.6 miles
- Median distance to nearest = 1.4 miles
- Transit connectivity isn’t high. This will likely bring scores down, though transit is lowest weighted at 25%.



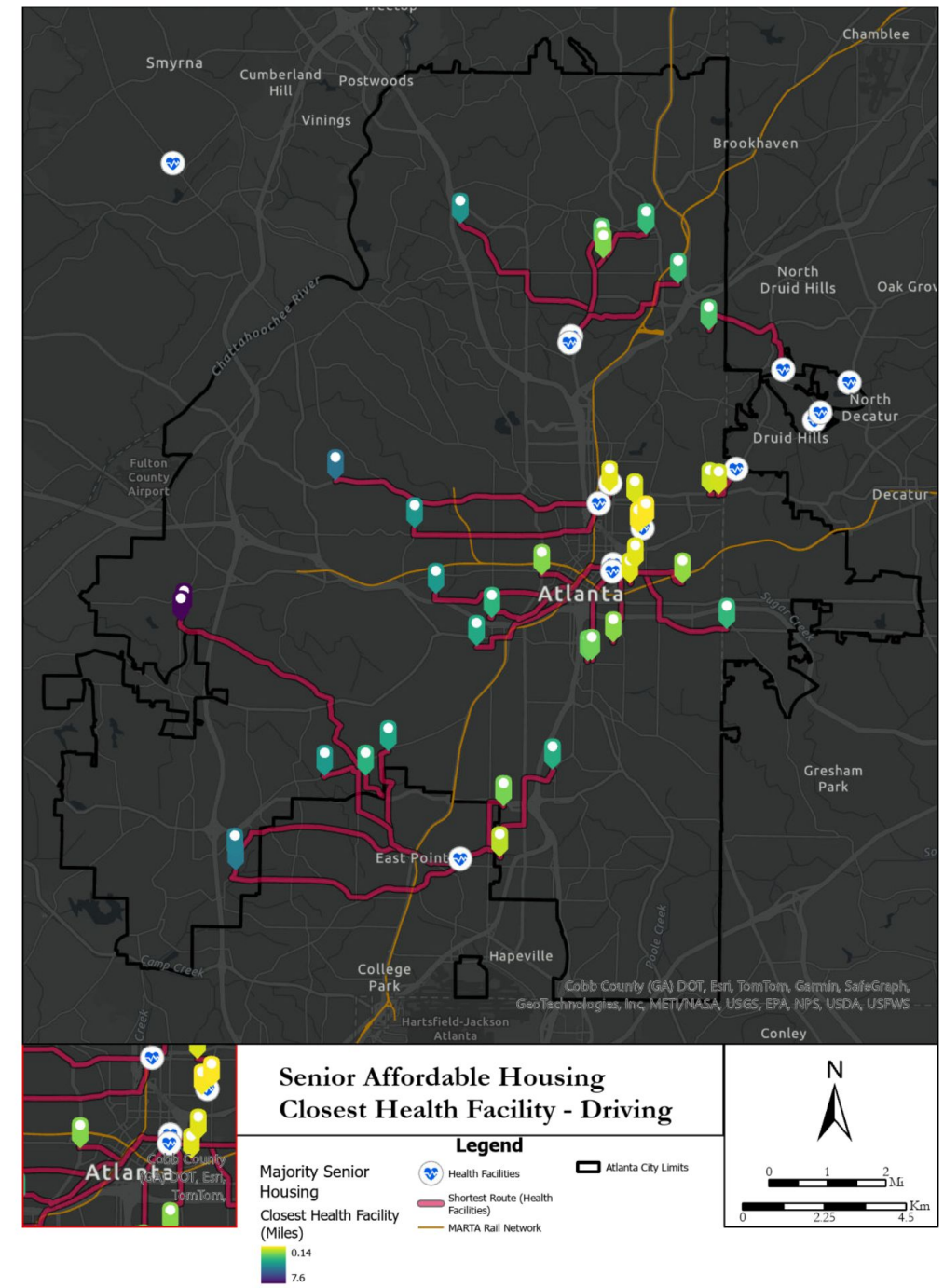
Closest Facility – Grocer

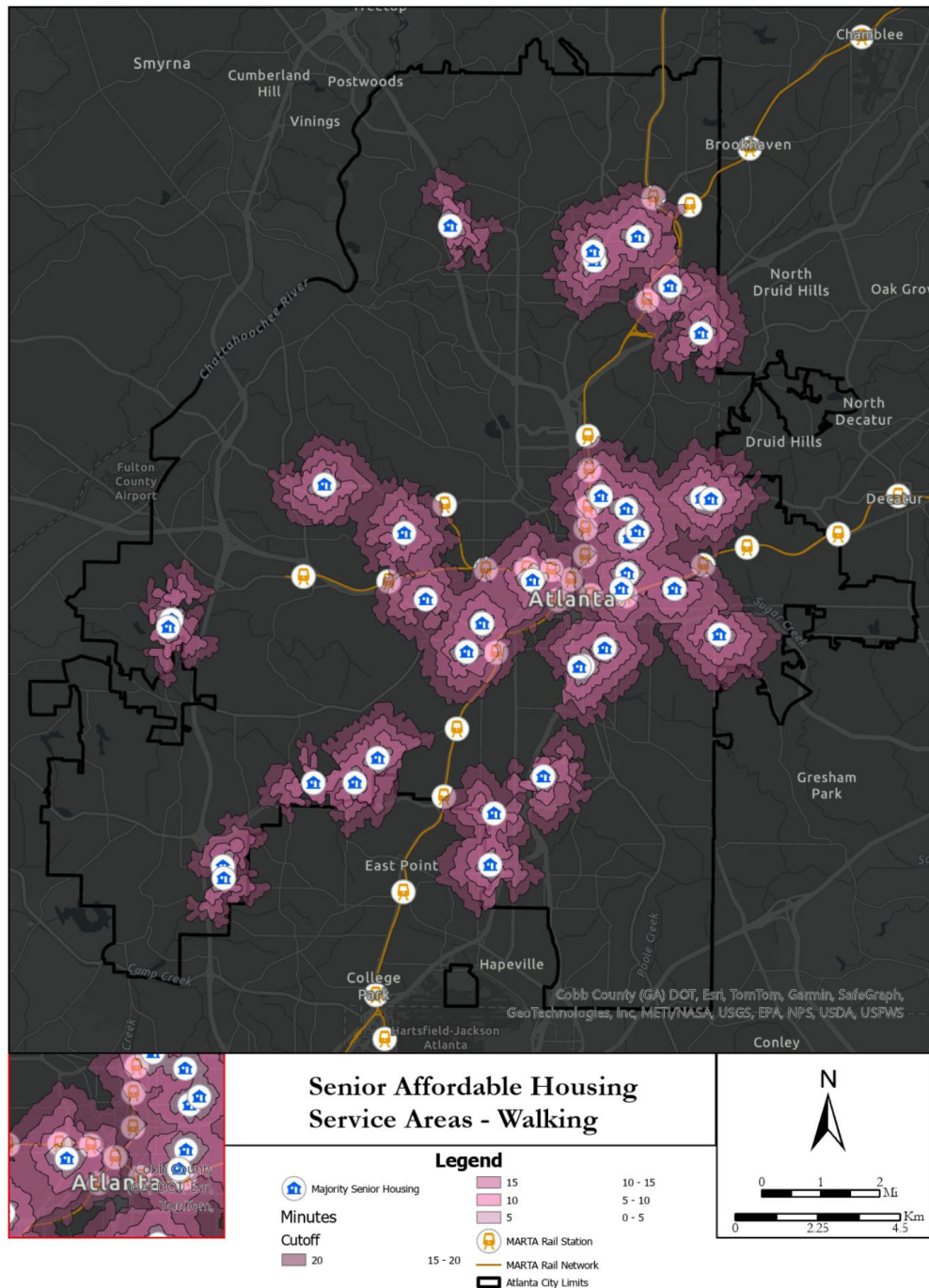
- Very short distances in northeast, complete opposite on southwest
- Average distance to nearest = 1.2 miles
- Median distance to nearest = 0.7 miles
- Large number of closer-by grocery stores on the northside are pulling these values up.
- These close distances are not the case for Southwest Atlanta.



Closest Facility - Health

- Technically used “drive” settings here for analysis due to max limit of distance on walk network.
- Average distance to nearest = 2.6 miles
- Median distance to nearest = 2.5 miles
- A large number of distant facilities are pulling this higher.
- This means that while some are very close, it’s not the vast majority of the population.



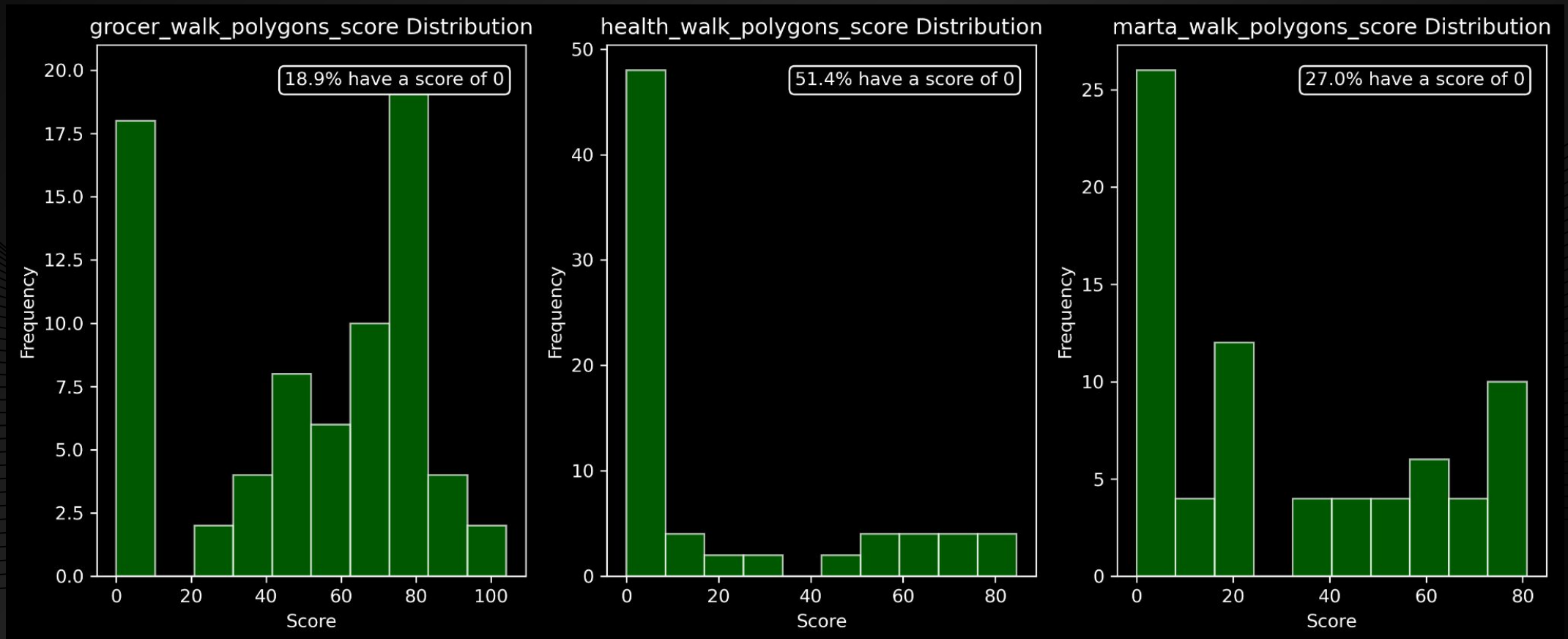


Methods for Accessibility Index, Part 1

1. **Service Areas** of 5, 10, 15, and 20 minutes walking for each site type (housing as the base, while dimensions were MARTA, grocery, health).
2. **Weighting** based on distance (i.e. 0–5 minutes = 4, 15–20 minutes = 1).
3. **% overlap** of each polygon at each distance band, creating distance decay.
4. **Sum, Min–Max Normalization**, placing values between 0 and 1.
5. **Dimension weighting** (Health = 45%, Grocer = 35%, MARTA = 25%)

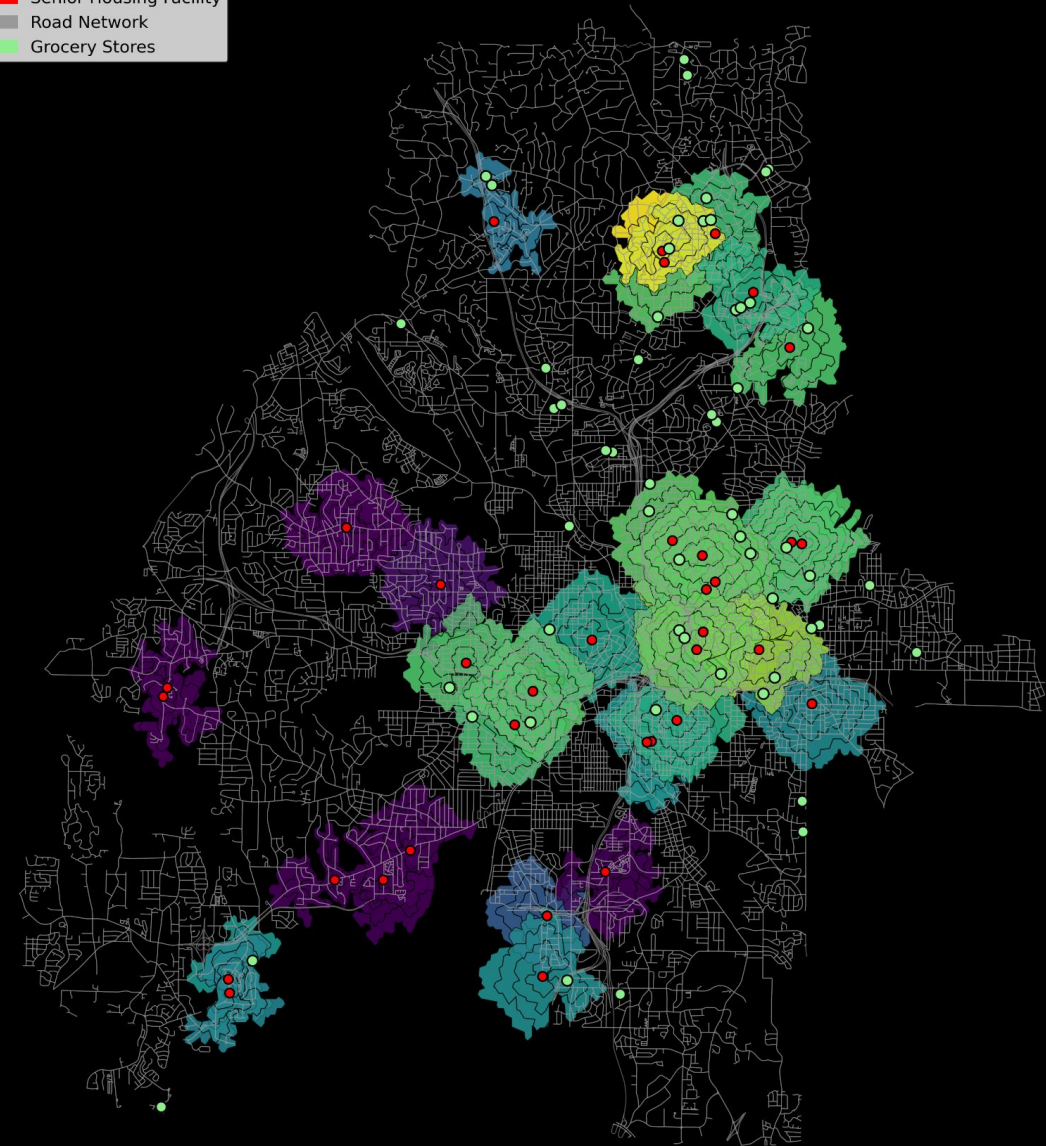
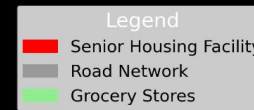
Methods for Accessibility Index, Part 2

A score of 0 means it was a housing facility not in a service area. The values on the left are out of 37 total facilities across 4 different time bands.



Dimension Scores – Housing in Red

- Grocery: Highest co-clustering in Buckhead area, 30%
 - Grocery stores in Green
 - Mean: 0.49
 - Median: 0.57



Grocery Access Score:
Mean Score by Facility: 0.49
Median Score by Facility: 0.57

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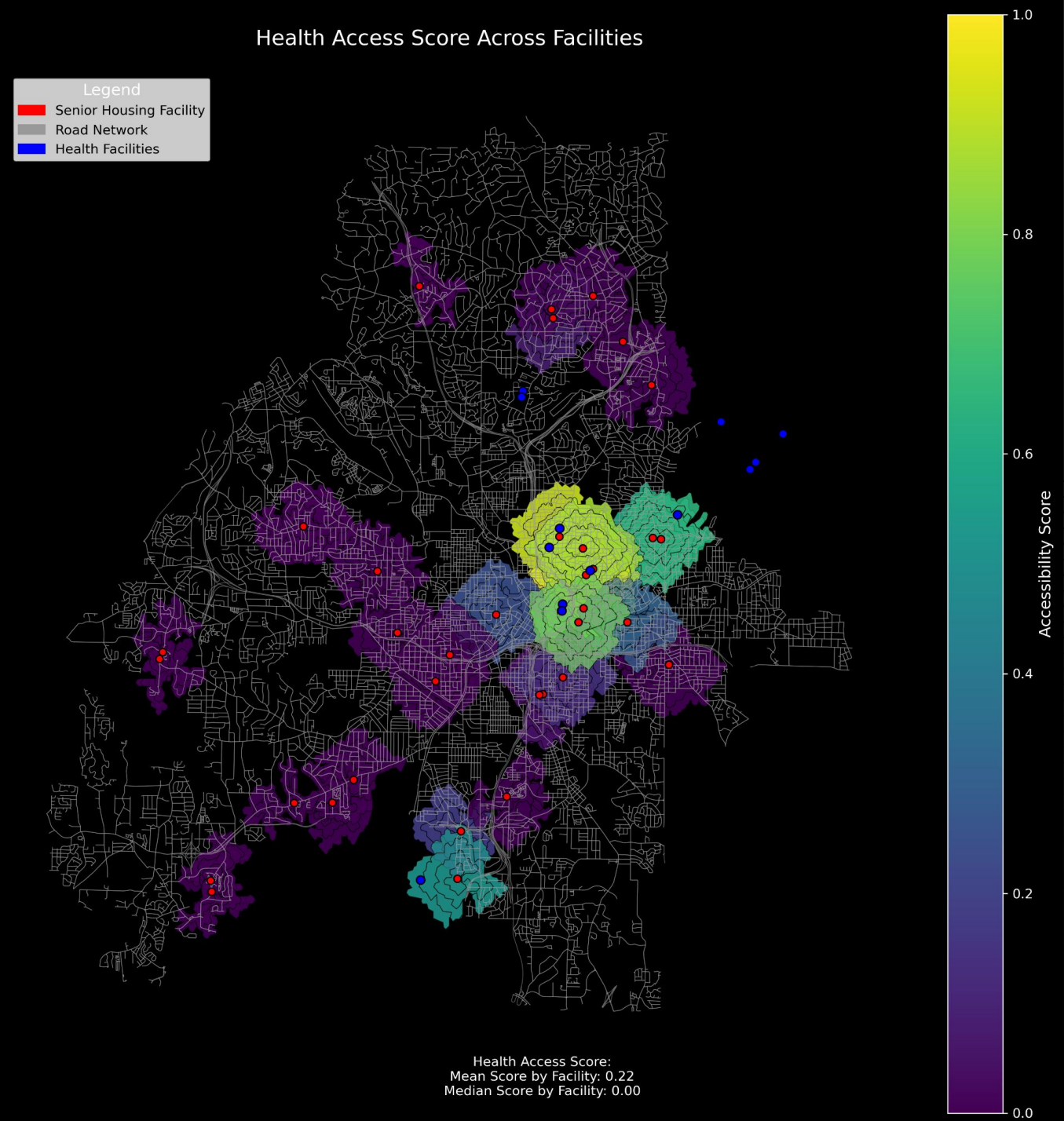
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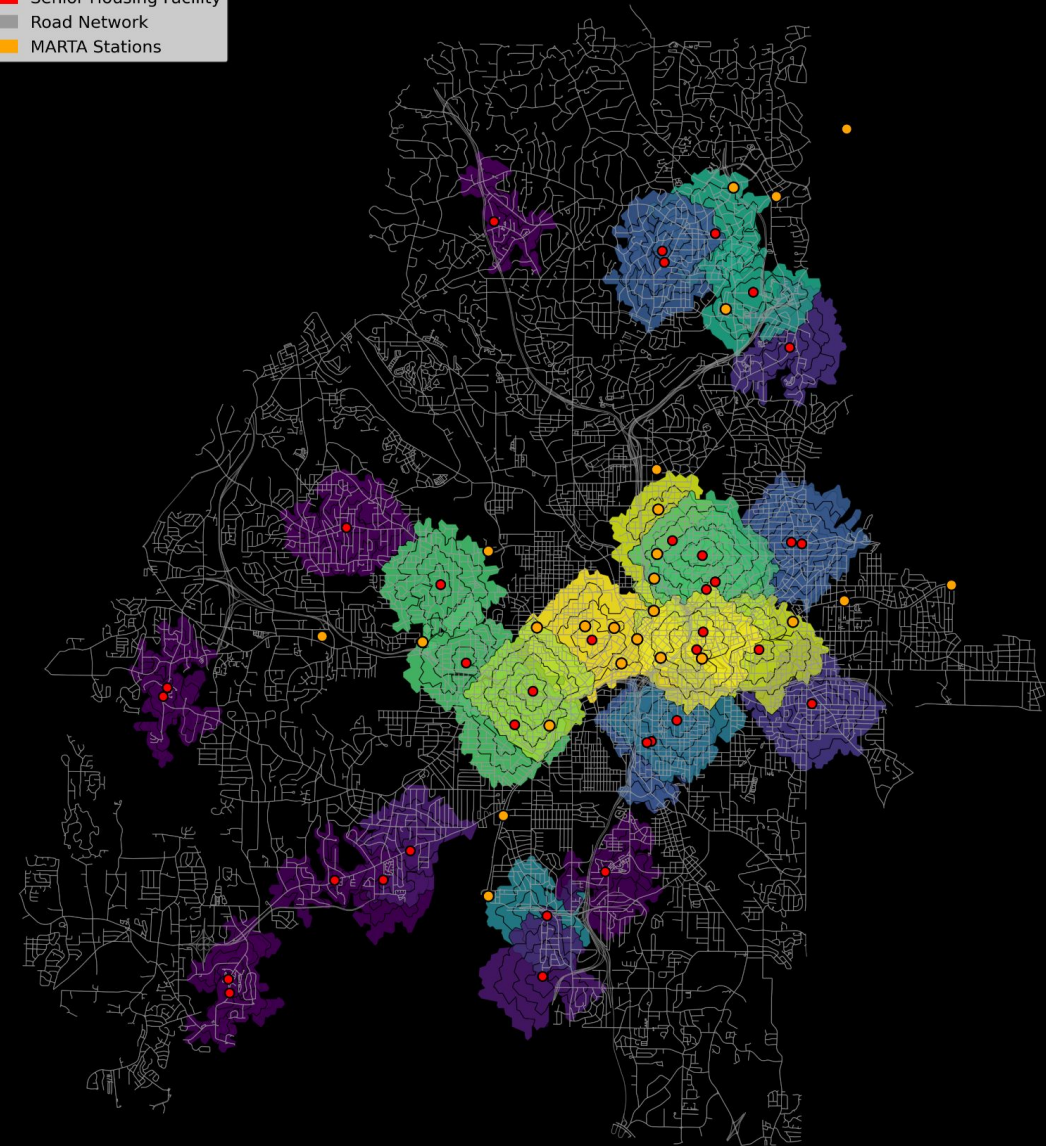
Dimension Scores – Housing in Red

- Grocery: Highest co-clustering in Buckhead area, 35%
- Health: Highest co-clustering in north-central area, around Midtown and O4W, 45%
 - Health facilities in Blue
 - Mean: 0.22
 - Median: 0.00



Dimension Scores – Housing in Red

- Grocery: Highest co-clustering in Buckhead area, 35%
- Health: Highest co-clustering in north-central area, around Midtown and O4W, 45%
- MARTA: Highest co-clustering in center corridor / just east and west of it, 25%
 - Stations in Orange
 - Mean: 0.38
 - Median: 0.27



MARTA Access Score:
Mean Score by Facility: 0.38
Median Score by Facility: 0.27

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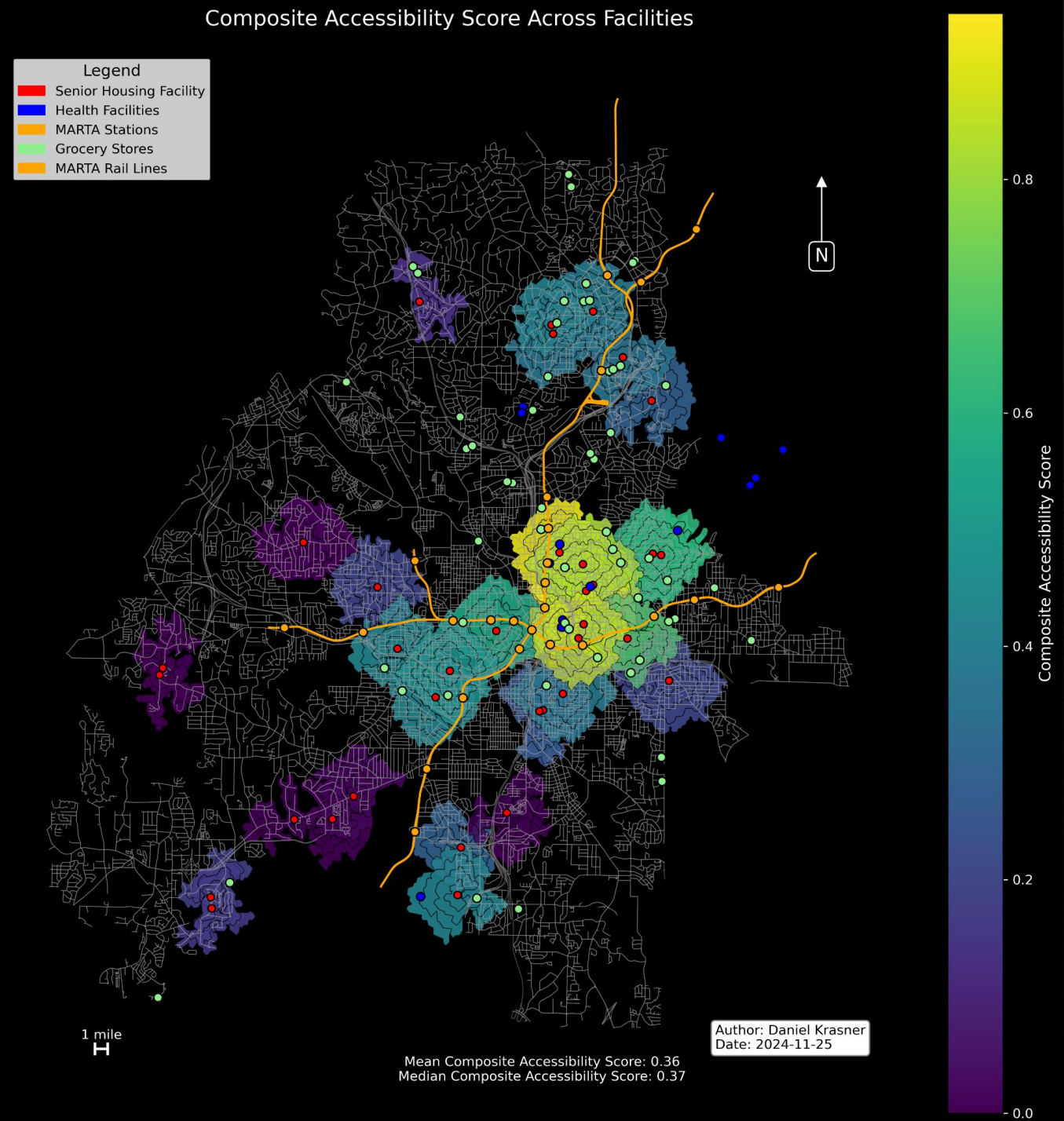
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Dimension Scores – Housing in Red

- Grocery: Highest co-clustering in Buckhead area, 35%
- Health: Highest co-clustering in north-central area, around Midtown and O4W, 45%
- MARTA: Highest co-clustering in center corridor / just east and west of it, 25%

Composite Index Fixates Central /
Central North

- Mean: 0.36
- Median: 0.37



Additional Areas of Exploration

Demographic Disparities

The HUD data has specific demographic variables that could help gain a picture of tenants.

Slope

Route data can be joined with raster data to determine slope of shortest paths. Atlanta is a high altitude city with many hills.

Impervious Surfaces

Raster data on impervious surfaces can be used to determine tree coverage or overrepresentation of surfaces that retain and reflect heat.

Greenspace

Polygon data can be added as a 4th variable to determine accessibility / distance to greenspace.

Dynamic Transit Model

Weighting of model could be conditional upon meeting a certain threshold. For example, MARTA is 25% unless the facility is high proximity to it AND low proximity to other variables, then it inflates in importance.

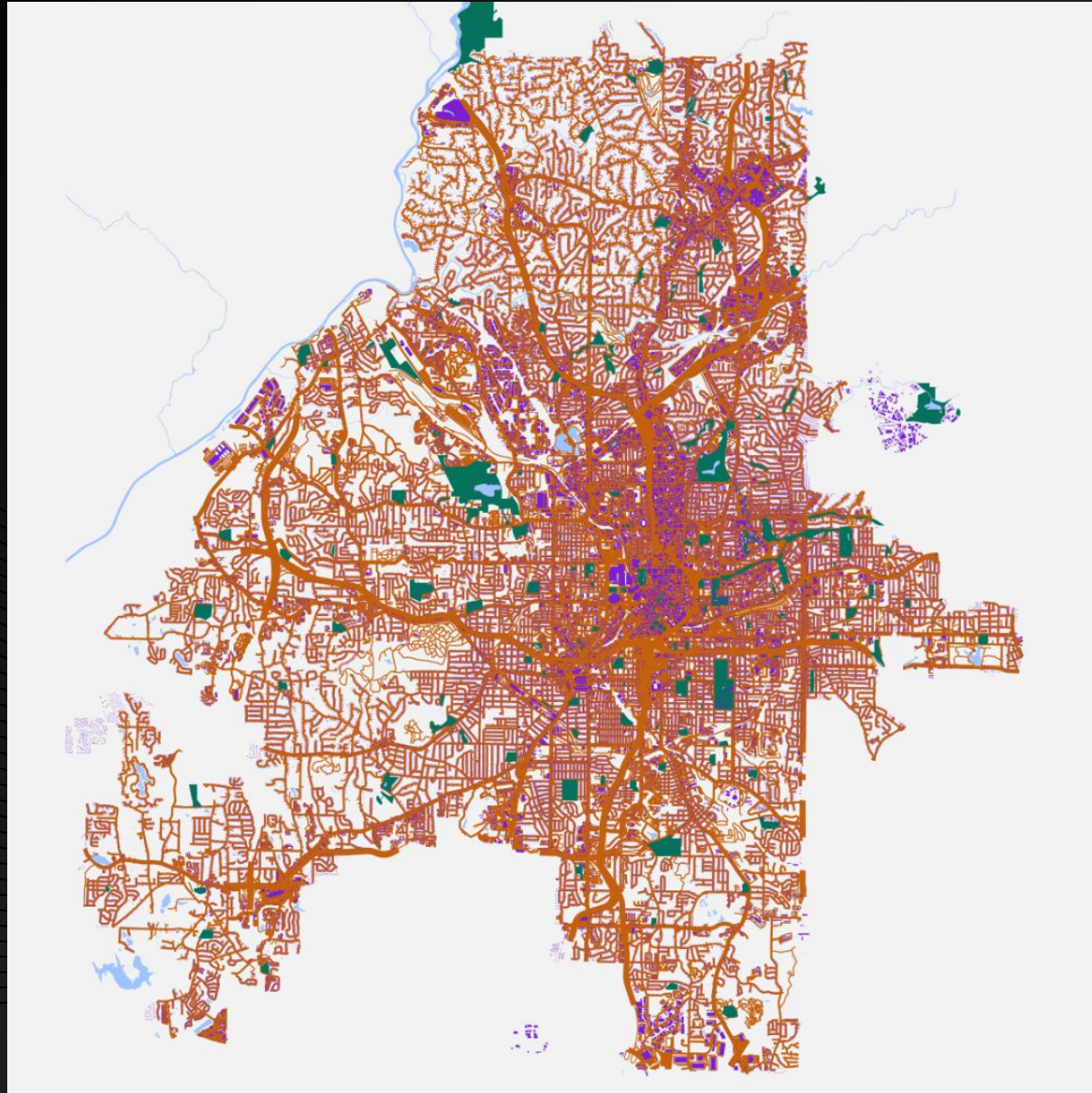
Sources

Quantitative

- HUD, Picture of Subsidized Households, 2023 – Based on Census 2020 geographies, Project level, https://www.huduser.gov/portal/datasets/pictures/files/PROJECT_2023_2020census.xlsx
- OSM (walking network, grocery stores, and MARTA Rail)

Qualitative

- As Metro Atlanta's Area Agency on Aging, ARC Serves Needs of Older Residents, Atlanta Regional Commission, Feb 24, 2022
- Aging Snapshot, Regional Trends in the Metro Atlanta Older Adult Population, Maria Sotnikova, Atlanta Regional Commission, May 2021
- 'Aging in place' options scarce in metro Atlanta, Georgia Public Broadcasting, Cathy Cobbs, May 1, 2023
- Report: Georgia among the worst places for older adults to live, The Atlanta Journal-Constitution, Anagha Ramakrishnan, Aug 3, 2022
- New Georgia data: There are more of us, we're getting older, The Current, Rebecca Grapevine, July 5, 2022



Thank you!

Do you have any questions?